

Solar Photovoltaic Power Forecasting
A Machine Learning Regression Approach Using Irradiance and Temperature Data
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I. Introduction

The global transition toward renewable energy systems have positioned solar photovoltaics as one of the fastest-growing electricity generation techniques of the twenty-first century. Between 2010 and 2024, the installed global PV capacity grew from approximately 40 GW to over 1700 GW, representing a compound annual growth rate exceeding 25 percent. This rapid expansion has profoundly altered the operational characteristics of electricity grids, which were originally designed around controllable, dispatchable generation sources such as coal, gas, and nuclear plants. Unlike conventional generators, solar PV systems produce electricity in direct proportion to the availability of solar radiation – a quantity that is inherently variable, intermittent, and weather-dependent. The output of solar installation can vary from near-maximum capacity to zero within minutes when cloud banks obscure the sun, creating sharp ramps in generation that grid operators must accommodate through fast-response reserves or energy storage. Accurate forecasting of PV generation is therefore not merely an academic exercise, but rather a practical necessity for maintaining grid stability, minimizing the cost of ancillary services, and enabling higher penetrations of renewable energy without compromising supply reliability.

Machine learning (ML) has emerged as the leading methodological paradigm for solar power forecasting, surpassing traditional physical and statistical approaches in accuracy and adaptability. Physical (numerical weather prediction) models are computationally expensive and require detailed atmospheric data; statistical models such as ARIMA perform poorly in the presence of the strong non-stationarities introduced by the day-night cycle and seasonal variation. By contrast, data-driven ML models can implicitly learn the complex, non-linear relationships between meteorological variables and PV output directly from historical observations, without requiring an explicit mathematical model of the physical generation process.

This research paper investigates the application of supervised ML regression techniques to a real-world hourly solar dataset collected over four years (2016 - 2019) at a single PV installation. The dataset contains 31,610 timestamped records of ambient temperature, Global Horizontal Irradiance (GHI), and measured PV power output. The paper is structured to address nine fundamental research questions spanning dataset characterization, feature engineering, preprocessing methodology, model selection, and performance evaluation, providing a complete and self-contained treatment of the solar PV forecasting problem for researchers.

1.1 Research Objectives

This paper pursues the following primary research objectives:

- How to characterize statistical properties and temporal patterns of a multi-year solar PV dataset.
- Identify the key physical and statistical drivers of PV power generation from the available features.
- Design and justify a complete data processing and feature engineering pipeline.
- Evaluate the sustainability of multiple ML algorithms – from simple linear models to ensemble methods – for the regression task.
- Recommend an evaluation framework with appropriate metrics for assessing forecast accuracy.
- Provide detailed, evidence-based answers to nine conceptual research questions on solar forecasting methodology

1.2 Paper Organization

The remainder of this paper is organized as follows:

- Section 2 provides the scientific and technical background on solar energy, PV systems, and GHI.
- Section 3 reviews related literature.
- Section 4 explains core machine learning concepts relevant to the research problem.
- Section 5 describes the dataset in detail
- Section 6 presents the preprocessing methodology.
- Section 7 will address all nine research questions.

- Section 8 shall outline the recommended ML pipeline and model comparison
- Section 9 will discuss broader implications.
- Section 10 will conclude the paper, followed by References.

II. Scientific & Technical Background

2.1 Photovoltaic Effect

The photovoltaic (PV) effect, first observed by Edmond Becquerel in 1839, refers to the generation of an electric voltage when certain materials are exposed to light. In a modern silicon solar cell, photons from incident sunlight excite electrons in the p-n junction semiconductor structure, causing them to flow as direct current (DC) electricity. An inverter converts this DC output into alternating current (AC) suitable for grid connection. The power output P on an ideal PV cell is given by:

$$P = \eta \times A \times G \times [1 - \beta(T - T_{ref})]$$

Where:

- P is the output of the solar panel
- η is the efficiency of the PV module
- A is the panel area (m^2)
- G is the incident irradiance (W/m^2)
- β is the temperature coefficient (typically 0.004 per $^{\circ}C$ for silicon)
- T is the operating temperature ($^{\circ}C$)
- T_{ref} is the standard test temperature ($25^{\circ}C$).

This formula reveals the two primary weather-driven inputs to the PV generator: irradiance and temperature.

2.2 Solar Irradiance: GHI, DNI, and DHI

Solar irradiance quantifies the power of solar radiation per unit area, measured in watts per square metre (W/m^2).

Three standard decompositions are widely used in solar energy research:

- Global Horizontal Irradiance (GHI): The total solar radiation received on a horizontal surface, encompassing both direct and diffuse components. GHI is most commonly measured and reported irradiance quantity, and it is the primary feature used in this study.
- Direct Normal Irradiance (DNI): The beam irradiance received on a surface perpendicular to the sun, relevant for concentrated solar power (CSP) and high-tilt PV installations.
- Diffuse Horizontal Irradiance (DHI): The scattered radiation received from the sky dome (excluding the direct solar disc), dominant on overcast days.

The decomposition relationship is: $GHI = DNI \times \cos(\theta_z) + DHI$, where θ_z is the solar zenith angle. GHI varies diurnally (zero at night, peak at solar noon) and seasonally (high summer values at most latitudes due to a smaller zenith angle). In the dataset used in this study, the GHI ranges from 0 to 1,050.3 W/m^2 .

2.3 The Daily & Seasonal Solar Cycle

The Earth's rotation about its axis and its orbit around the sun produce two superimposed periodicity patterns in solar irradiance that directly drive PV generators: the diurnal (24-hour) cycle and the annual (seasonal) cycle. The diurnal cycle produces irradiance during night hours, a rapid increase after sunrise, a peak around local solar noon, and a symmetric decline until sunset. The duration of the daylight period varies from approximately 10 to 14 hours depending on the season and latitude.

The annual cycle introduces seasonal variation in both peak GHI (higher in summer when the sun is higher in the sky) and total daily generator hours. These two nested periodicities are the reason why time-based features are extracted from the DateTime column – such as hour of the day and month – are such powerful predictors of PV power output, even beyond their direct relationship through GHI.

2.4 Temperature Effects on PV Performance

While GHI is the dominant determinant of PV output, temperature has a well-documented secondary effect on panel efficiency. Silicon solar cells exhibit a negative temperature coefficient: as cell temperature rises above the standard test condition of 25°C, the open-circuit voltage decreases, causing a reduction in power output of approximately 0.3-0.5% per degree Celsius for typical polycrystalline panels. In the dataset for this particular study, ambient temperatures reach 47.6°C, which could reduce panel efficiency by up to 10% relative to standard conditions. This explains the positive but weaker correlation ($r = 0.365$) between temperature and PV power: on a seasonal basis, warm months also have higher GHI, creating a confounded positive correlation, while at the panel level, higher temperatures actually reduce efficiency.

III. Literature Review

3.1 Evolution of Solar Forecasting Methods

Solar power forecasting has evolved through three broad generations of methods: physical models, statistical time-series models, and machine learning models. Each generation has addressed the shortcomings of its predecessors while introducing new challenges.

Physical models, based on Numerical Weather Prediction (NWP) systems, simulate atmospheric dynamics to forecast irradiance and temperature. While physically interpretable, NWP models require substantial computational infrastructure, have limited spatial resolution, and typically achieve useful forecast horizons of 6-48 hours. Studies by Lorenz et al (2011) demonstrated that NWP-based forecasts achieve day-ahead accuracy of approximately 15-25% in terms of relative RMSE for individual sites, with performance degrading significantly for sub-hourly horizons.

Statistical models, including ARIMA (Auto-Regressive Integrated Moving Average) and exponential smoothing, exploit temporal autocorrelation in historical generation data. However, these approaches struggle with the non-stationarity of the solar time series – the clear diurnal and season periodicity requires careful seasonal differencing – and they cannot generalize across sites or incorporate meteorological covariates without significant modification.

Machine learning approaches have increasingly dominated the literature since about 2012. Gradient boosting methods, artificial neural networks, and support vector machines have all been applied successfully to solar forecasting tasks. A systematic review by Wang et al. (2019) found that ensemble three-based models – particularly Random Forest and XGBoost – consistently achieved the highest accuracy on hour-ahead and day-ahead forecasting benchmarks when trained on irradiance and temperature features.

3.2 Key Studies in ML-Based Solar Forecasting

Antonanzas et al. (2016) benchmarked 15 machine learning algorithms on a one-year hourly solar generation dataset. Random Forest achieved the lowest RMSE (9.2 kW) compared to Artificial Neural Networks (11.4 kW) and Support Vector Regression (12.8 kW), while Linear Regression served as the baseline (18.3 kW). The study also identified GHI as the single most important feature, accounting for over 80% of the total feature importance in the Random Forest model.

Chen et al (2021) introduced a hybrid model combining XGBoost with LSTM for multi-step solar forecasting, demonstrating R^2 values of 0.981 on 1-hour-ahead predictions and 0.943 on 6-hour-ahead predictions. Their work highlighted the complementary strengths of gradient boosting (strong short-term accuracy) and LSTM (temporal dependency modelling for longer horizons).

Huang and Xu (2020) investigated feature engineering strategies for solar forecasting, demonstrating that including hour-of-day and month features as explicit inputs to gradient boosting models improved R^2 by 0.03 - 0.07 compared to models trained on raw meteorological variables alone. This finding directly informs the feature engineering strategy recommended in this paper.

3.3 Research Gap Addressed

While the ML solar forecasting literature is extensive, most studies focus narrowly on model accuracy comparisons without providing a comprehensive conceptual grounding suitable for scientists new to the domain. This paper addresses this gap by integrating background scientific knowledge, rigorous dataset analysis, and a complete methodological guide within a single coherent framework anchored to a real four-year dataset.

IV. CORE MACHINE LEARNING CONCEPTS

4.1 Supervised Learning and Regression

Machine learning encompasses three principal paradigms: supervised learning, unsupervised learning, and reinforcement learning. Solar power forecasting is a supervised learning problem because a labelled training set – historical records of inputs (temperature, GHI) and corresponding outputs (PV power) – is available from which a model can learn the input-to-output mapping.

Within supervised learning, two task types exist: classification (predicting discrete categories) and regression (predicting continuous numerical values). PV power output is a continuous variable measured in kilowatts, making this unambiguously a regression problem. The goal is to learn a function f such that:

$$PV\ Power_t = f(GHI_t, Temp_t, Hour_t, Month_t, ...) + \varepsilon_t$$

Where:

- $PV\ Power_t$ is the solar power output at time t .
- GHI_t is the global horizontal irradiance at time t .
- $Temp_t$ is the temperature at time t .
- $Hour_t$ is the hour of the day.
- $Month_t$ is the month or seasonal indicator
- $f(\cdot)$ is the predictive function learned by the model being:
 - Linear regression
 - Random forest
 - XGBoost
 - Neural Network
 - LSTM
- ε_t is the random error and or the irreducible noise (e.g soiling, partial shading, inverter losses) that no model can predict from the available features.

In other words, this equation is essential stating “Solar power output depends on environmental and temporal features, while accounting for some unavoidable prediction error”.

In machine learning terms: $Target\ Variable = Model(Features) + Error$, where “Target” is the PV Power, “Features” is weather + time variables, and “Error” is uncertainty.

4.2 The Bias-Variance Tradeoff

A central concept in supervised learning is the bias-variance tradeoff, which governs the generalization behaviour of predictive models. Bias refers to systemic prediction error arising from overly simplistic model assumptions; variance refers to sensitivity to fluctuations in the training data. The expected test error of any model can be decomposed as:

$$Expected\ Error = Bias^2 + Variance + Irreducible\ Noise$$

A simple model such as Linear Regression has a high bias (since it cannot capture non-linear relationships between GHI and PV Power during cloudy periods) but low variance, as its predictions are stable across different training samples). A deep neural network has low bias but high variance – it can fit the training data almost perfectly but may generalize poorly to new observations. Ensemble methods such as Random Forest and XGBoost are explicitly designed to balance this tradeoff by averaging or sequentially correcting many weak learners.

4.3 Overfitting, Underfitting, and Generalization

Overfitting occurs when a model learns the specific noise and idiosyncrasies of the training data rather than the underlying generative process. An overfitting model achieves very low training error but high test error. Underfitting, conversely, occurs when the model is too simple to capture the true relationship – producing high error on both training and test sets.

For the solar forecasting dataset studied here, overfitting is a realistic concern despite the large sample size (31,610 observations), because models with many parameters (deep trees, large neural networks) can memorize patterns specific to the 2016-2018 training period that do not transfer to 2019. Regularization techniques such as L1/L2 penalties, tree depth limits, minimum leaf sample requirements, and dropout in neural networks – are the primary defences against overfitting.

4.4 Cross-Validation in Time-Series Contexts

Standard k-fold cross-validation randomly partitions the dataset into k subsets and rotates through them as train and test sets. For time-series data, random partitioning is invalid because it allows future data to appear in the training set (data leakage), producing unrealistically optimistic performance estimates. The correct approach is TimeSeriesSplit cross-validation, in which folds are always ordered chronologically: all training data precedes the validation data in time. This mirrors the real-world forecasting scenario, where models trained on past data are evaluated on future data.

4.5 Key Evaluation Metrics for Regression

Three standard metrics are used to evaluate regression model performance in the solar forecasting literature:

- **Mean Absolute Error (MAE):** Computed as the arithmetic mean of absolute differences between predicted and actual values. MAE has the same units as the target variable (kW), is easy to interpret, and is robust to outliers. It does not penalise large individual errors disproportionately.
- **Root Mean Squared Error (RMSE):** Computed as the square root of the mean squared error. RMSE penalizes large individual errors more heavily than MAE due to the squaring operation, making it sensitive to occasional severe mispredictions – an important property in grid operations where large errors are costlier than smaller ones.
- **R² (Coefficient of Determination):** measures the proportion of variance in the target variable explained by the model. R² = 1.0 indicates a perfect model; R² = 0 indicates a model no better than predicting the mean; negative values indicate a model worse than the mean predictor.

4.6 Principal Machine Learning Algorithms

4.6.1 Linear Regression

Linear Regression assumes a linear relationship between the input features X and the target variable y :

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n.$$

Parameters β are estimated by minimizing the residual sum of squares

(ordinary least squares). Despite its simplicity, linear regression provides a crucial baseline: it is the minimum performance any more complex model must exceed to justify its additional complexity. For this dataset, linear regression with GHI as the sole predictor already achieves a respectable R² given the near-perfect linear correlation ($r = 0.0095$), but it fails to capture non-linear effects during partially cloudy periods.

4.6.2 Decision Trees and Random Forests

A decision tree recursively partitions the feature space into axis aligned rectangular regions, fitting a constant prediction within each leaf node. While interpretable, single decision trees overfit easily. Random Forest addresses this by training an ensemble of B-Trees, each on a bootstrap sample of the training data and random subset of features of each split. Predictions are averaged across trees (bagging). The dual randomization – bootstrap samples and feature subsampling – reduces variance without significantly increasing bias, producing a model that is

robust to outliers, handles non-linearity naturally, and is relatively insensitive to hyperparameter choices. Random Forest is considered the workhorse of tabular ML forecasting and is particularly well-suited to the solar dataset studied here.

4.6.3 Gradient Boosting and XGBoost

Gradient Boosting builds an additive ensemble of weak learners (shallow trees) sequentially, each new tree fitting the negative gradient of the loss of the function of its predecessors – effectively correcting the residual errors from prior interactions. XGBoost (Extreme Gradient Boosting) extends this framework with second-order gradient statistics, L1/L2 regularization on leaf weights, and efficient handling of sparse data and missing values. These innovations make XGBoost consistently the top-performing algorithm on structured tabular datasets across machine learning competitions and real-world benchmarks. For the solar forecasting problem, XGBoost’s ability to model complex feature interactions – such as the non-linear interaction between GHI level and time of day – is particularly valuable.

4.6.4 Long Short-Term Memory Networks (LSTM)

LSTMs are a class of recurrent neural network (RNN) specifically designed to learn long-range temporal dependencies in sequential data. Unlike feedforward models that treat each observation independently, LSTMs maintain a hidden state that carries information across timesteps through gated memory cells (input gate, forget gate, output gate). For solar forecasting, LSTMs can, in principle, capture the temporal autocorrelation of cloud cover patterns and the inertia of the thermal state of the panels. In practice, however, for single-step (1 hour ahead) forecasting, well-tuned gradient boosting models with lagged features match or exceed LSTM performance at far lower computational costs. LSTMs become the preferred choice when multi-step (24 hour ahead) probabilistic forecasting is required.

V. DATASET DESCRIPTION & EXPLANATORY ANALYSIS

5.1 Dataset Overview

The dataset used in this study comprises hourly observations from a solar PV installation collected over four complete calendar years: January 1, 2016 to December 31, 2019. In total, 31,609 valid timestamped records are present, representing near-continuous measurement at 1-hour resolution with only 15-16 missing values (<0.05% of the dataset). Each record contains three numeric measurements: ambient air temperature (°C). Global Horizontal Irradiance (W/m²), and PV power output (kW).

5.2 Column Definitions

Table 5.2.1: this table summarizes the variables included in the dataset and their relevance to the research.

Column	Units	Range	Description
DateTime	Timestamp per hour	2016-01-01 01:00 - 2019-12-31 23:00	Date and time of the observation at 1-hour intervals.
Temperature	°C	3.4 - 47.6	Ambient outdoor air temperature at the measurement station.
GHI	W/m ²	0 - 1,050.3	Global Horizontal Irradiance: total radiation on a horizontal surface.
PV Power	kW	0 - 195.75	Target variable: measurement AC power output of the PV system at each hour.

5.3 Descriptive Statistics

Table 5.3.1: this table analyzes the data collected for each variable (calculations can be found in the glossary).

Variable	Count	Mean	Standard Deviation	Min	25%	75%	Max
Temperature	31,593	27.14	7.40	3.4	21.1	33.4	47.6
GHI (W/m ²)	31,594	230.09	305.97	0	0	470.1	1,050.3
PV Power (kW)	31,593	44.03	58.10	0	0	90.71	195.75

The 25th percentile of both GHI and PV Power is zero, confirming that more than 25% of all hours are night-time observations. The high standard deviation of GHI (305.97 W/m² on a mean 230.09) reflects the bimodal nature of the distribution: the large mass of zero values (night) and a right-skewed distribution of positive values during daylight hours.

5.4 Correlation Analysis

The Pearson correlation coefficients between the three numeric variables were computed on the full dataset. The results reveal a strikingly strong relationship between GHI and PV Power:

Feature Pair	Pearson r	Interpretation
GHI ↔ PV Power	0.9995	Near-perfect linear
Temperature ↔ PV Power	0.3648	Moderate positive
Temperature ↔ GHI	0.3611	Moderate positive

The correlation between GHI and PV Power ($r=0.9995$) is extraordinarily high for a real-world physical system, reflecting the direct proportionality between incoming solar radiation and electrical output described by the PV conversion equation. The Temperature ↔ PV Power correlation ($r = 0.3648$) is confounded by the seasonal co-variation of temperature and GHI: summer months have both higher temperatures and more sunlight, creating an apparent positive association even though the direct effect of temperature on panel efficiency is negative.

5.5 Temporal Pattern Analysis

Analysis of PV power by hour of day confirms the expected bell-shaped diurnal profile. Generation is zero between approximately 19:00 and 06:00 (14,428 zero-GHI records), rises steeply from sunrise (~06:30), reaches its maximum around 11:00 - 13:00 (peak values of 195.75 kW), and declines symmetrically until sunset (~18:30). Monthly analysis across the four-year span reveals clear seasonal variation, with June - August producing the highest mean hourly generation and December – January the lowest, consistent with the latitude of the installation.

VI. DATA PREPROCESSING METHODOLOGY

6.1 Missing Value Treatment

Fifteen to sixteen observations are missing across the three numeric columns, representing approximately 0.05% of the dataset. Three options are available: listwise deletion (removing incomplete rows), forward-fill imputation (using the preceding hour's value), or linear interpolation between adjacent valid timestamps. Given the negligible volume, listwise deletion is recommended as the simplest and most conservative approach, avoiding any potential distortion of the time series through imputed values.

6.2 DateTime Parsing and Feature Engineering

The raw DateTime column contains a timestamp string that must be converted to a datetime object before feature extraction. From each timestamp, the following features should be derived:

- **Hour (0-23):** encodes the position within the diurnal cycle. This is the single most informative engineered feature, as it directly predicts the expected GHI range at any point in the day.
- **Month (1-12):** captures seasonal variation in peak GHI and daylight duration. Together with Hour, it encodes the expected solar geometry at each observation.
- **DayOfWeek(0-6):** Has minimal direct relevance for PV generation (solar output is not affected by the calendar), but may carry indirect signals related to temperature patterns.
- **Season (0-3):** Derived from Month as a coarser seasonal indicator: Winter (DJF), Spring (MAM), Summer (JJA), Autumn (SON).
- **IsDaytime (0/1):** A binary flag derived from $GHI > 0$. Explicitly marks night-time observations, helping tree-based models avoid unnecessary splits on other features at night.

6.3 Outlier Detection Treatment

Physical bounds for the three variables are well-defined by the underlying physics. Any observation with $GHI > 1,400 \text{ W/m}^2$ (the solar constant at Earth's surface), Temperature outside the range $[-20^\circ\text{C}, 60^\circ\text{C}]$, or negative PV Power should be flagged as erroneous and removed. For statistical outlier detection within the physical bounds, the interquartile range (IQR) method can be applied: values more than $3 \times \text{IQR}$ above the 75th percentile or below the 25th percentile are flagged for review. The zero-inflation of GHI and PV Power (night-time values) must be accounted for by applying outlier detection only to daytime observations.

6.4 Feature Scaling

Feature scaling normalizes the numerical range of input features to a common scale, preventing features with larger magnitudes from dominating distance-based or gradient-based algorithms. Two approaches are standard:

- **StandardScalar:** Transforms each feature to zero mean and unit variance: $x' = (x - \mu) / \sigma$.
 - Appropriate when features are approximately normally distributed.
- **MinMaxScalar:** Scales each feature to the $[0,1]$ range: $x' = (x - x_{\min}) / (x_{\max} - x_{\min})$.
 - Appropriate when a bounded range is required (i.e for neural network activation functions).

Critically, the scalar must be fitted exclusively on training data and then applied to both training and test data. Fitting the scalar on the full dataset before splitting constitutes data leakage, as test set statistics would influence the training set representation.

6.5 Train/Test Split Strategy

For time-series data, the split must be chronological to prevent data leakage. The recommended split uses 2016-2018 (approximately 75% of the data) as the training set and 2019 (approx. 25%) as the held-out test set. This mimics the operational deployment scenario where a model trained on historical data is used to forecast the following year. Random shuffling before splitting is strictly contraindicated for time-series regression.

VII. RESEARCH QUESTIONS & ANALYSIS

This section provides detailed, evidence-based responses to the nine research questions posted for this assignment. Each response draws on the dataset analysis in Section 5, the background in Sections 2-4, and the preprocessing methodology in Section 6.

RQ1 - What is the main objective of this dataset and what are we trying to predict?

The primary objective of this dataset is to enable the development of a supervised machine learning regression model capable of forecasting solar PV power output (kW) from contemporaneous and historical weather measurements. Specifically, we seek to learn a function $f(\text{GHI}, \text{Temperature}, \text{Hour}, \text{Month}) \rightarrow \text{PV_Power}$ that generalizes from historical observations (2016-2018) to unseen future data (2019) with the lowest possible prediction error.

From an applied perspective, accurate short-term (1-hour-ahead) PV forecasts are used by grid operators to schedule dispatchable generation reserves, manage battery storage charge/discharge cycles, optimize energy trading in spot markets, and predict periods of excess renewable generation requiring curtailment or export. The economic value of a 10% improvement in forecast accuracy for a 1 MW installation has been estimated at USD 10,000 - 50,000 per year in avoided balancing costs, depending on the electricity market structure.

RQ2 - Which column is the target variable and why?

PV Power (kW) is the target (dependent) variable in this supervised regression task. It is designated as the target for three fundamental reasons. First, it represents the operational outcome of interest: the quantity that electricity grid operations, plant owners and energy traders need to forecast. Second, it is causally downstream of the weather variables - solar irradiance and temperature cause PV power generation through the photovoltaic conversion process, establishing the correct directional dependence for a regression model. Third, all other columns either serve as casual inputs (GHI, Temperature) or provide temporal context from which additional features can be derived (DateTime).

In the notation of supervised learning, PV Power is the label y in the training set $D = \{(x_i, y_i)\}_{i=1}^n$, where $x_i = (\text{GHI}_i, \text{Temp}_i, \text{Hour}_i, \text{Month}_i, \dots)$ is the feature vector for observation i . The model learns to map x to y by minimising a loss function (typically mean squared error) over the training set.

RQ3 - What are the input features in this dataset?

The dataset provides two directly measured meteorological features: Temperature (°C) and GHI (W/m²). Beyond these, a rich set of temporal features can be engineered from the DateTime column. The complete recommended feature set is presented in Table 7.1:

Feature	Source	Type	Predictive Rationale
GHI (W/m²)	Direct	Continuous	Primary driver of PV generation ($r = 0.9995$ with PV Power)
Temperature (°C)	Direct	Continuous	Affects panel efficiency; secondary predictor ($r = 0.365$)

Hour (0-23)	Engineered	Ordinal	Encodes intra-day solar position; key predictor of GHI range.
Month (1-12)	Engineered	Ordinal	Captures seasonal variation in daylight hours and solar angle.
Season (0-3)	Engineered	Categorical	Coarser season grouping; reduces cardinality for tree models.
IsDaytime (0/1)	Engineered	Binary	Explicit night indicator; prevents unnecessary splits at zero GHI
GHI_{t-1} (lag)	Engineered	Continuous	Previous hour's irradiance; captures cloud cover persistence.

RQ4. What does GHI represent and why is it important?

Global Horizontal Irradiance (GHI) is the total solar radiation power per unit area received on a horizontal surface at the ground level, expressed in watts per square metre (W/m^2). It combines direct beam radiation (DNI) and diffuse sky radiation (DHI): $GHI = DNI \times \cos(\theta_z) + DHI$, where θ_z is the solar zenith angle.

GHI is by far the most important feature in this dataset for the following evidence-based reasons. First, the Pearson correlation coefficient between GHI and PV Power is $r = 0.9995$, meaning that GHI alone explains 99.9% of the variance in pV output ($R^2 = \text{GHI-PV_Power correlation}^2 = 0.999$). No other single feature approaches this predictive power. Second, the mechanism is physically direct: PV panels convert photons (quantified by GHI) into electricity, so zero GHI necessarily implies zero PV output. This is confirmed in the dataset, where all 14,428 night-time records ($GHI=0$) have PV Power = 0. Thus, GHI naturally encodes both the diurnal cycle (zero at night, peak at noon) and the seasonal cycle (higher in summer), making it a proxy for multiple temporal features simultaneously.

RQ5. What pattern do you expect in PV power during day and night?

Based on solar physics and confirmed by the dataset, PV power follows a deterministic bell-shaped (sinusoidal) diurnal pattern superimposed on a stochastic noise component from cloud cover and atmospheric variability. The expected behavior by time period is:

Period	Approx. Hours	GHI Range	Expected PV Power
Deep night	21:00 - 05:30	0 W/m^2 (absolute)	0 kW (guaranteed)
Twilight / dawn	05:30 - 07:00	0 - 50 W/m^2	0-5 kW
Morning ramp	07:00 - 10:00	50 - 400 W/m^2	10 - 80 kW
Solar noon (peak)	10:00 - 14:00	400 - 1,050 W/m^2	80 - 196 kW

Afternoon decline	14:00 - 18:00	Decreasing	Decreasing symmetrically
Dusk / Sunset	18:00 - 19:30	0 - 50 W/m ²	0 - 5 kW

At the seasonal scale, summer months (June-August) produce both longer daylight windows and higher peak GHI values, resulting in significantly greater total daily energy. December and January produce the lowest peaks and shortest generation windows. Year-to-year variability is driven by interannual difference in cloud cover patterns, which are not predictable from the feature set alone.

RQ6. What preprocessing steps are required before applying a machine learning model?

A complete preprocessing pipeline for this dataset comprises the following six sequential steps, details in Section 6 and summarized here:

1. Missing Value Treatment: Identify and remove (or impute) the 16 rows with null values. Given the negligible volume (<0.05%), listwise deletion is recommended.
2. DateTime Parsing: Convert the DateTime string to a proper datetime type and sort the DataFrame chronologically. Extract Hour, Month, DayOfWeek, and Season as new feature columns. Create an IsDaytime binary indicator.
3. Outlier Screening: Apply physical bound checks ($GHI \leq 1,400 \text{ W/m}^2$, Temperature in $[-20, 60]^\circ\text{C}$, PV Power ≥ 0). For daytime observations only, apply the 3-IQR statistical outlier rule.
4. Feature Engineering: Create lagged features (GHI at t_1 and t_2) if temporal autocorrelation modelling is desired. Encode the Season feature as one-hot or ordinal integers.
5. Feature Scaling: Fit a StandardScalar on the training set features. Apply the fitted scalar to transform both training and test features. Never fit on the combined dataset.
6. Chronological Train/Test Split: Allocate 2016-2018 to training and 2019 to testing. Verify no timestamps from 2019 appear in the training set.

RQ7. Why is it important to split the dataset into training and testing sets?

A train/test split is the cornerstone of rigorous model evaluation in machine learning. Its purpose is to produce an unbiased estimate of how well the trained model will perform on unseen future data – the primary operational objective. Without a held-out test set, any performance metric computed on the training data reflects memorization ability, not generalization ability.

Two failure modes arise when a held-out test set is absent or improperly constructed. Overfitting occurs when the model learns the noise specific to the training period; it appears to perform excellently during training but fails on deployment data. Data leakage occurs when information from the future (test period) inadvertently influences the training process – for example, when a scalar is fitted on the combined dataset, or when cross-validation folds are not chronologically ordered. Both failure modes result in models that are optimistically evaluated but poorly performing in production.

The 80/20 chronological split recommended for this dataset (2016-2018 training, 2019 test) reflects standard practice in the solar forecasting literature. The test period spans all four seasons of a complete calendar year, ensuring that the evaluation captures seasonal variation and does not favour the model based on a particularly easy or difficult period.

RQ8. What evaluation metrics should be used to measure model performance?

Three metrics constitute the standard framework for regression in solar forecasting research and should be reported together to provide a complete picture of model accuracy:

Metric	Formula	Units	Characteristics
Mean Absolute Error (MAE)	$\frac{1}{n} \sum y_i - \hat{y}_i $	kW (same as target)	Interpretable; robust to outliers; does not penalise large errors disproportionately.
Root Mean Squared Error (RMSE)	$\sqrt{\frac{1}{n} (\sum y_i - \hat{y}_i)^2}$	kW (same as target)	Penalises large errors; sensitive to mispredictions during rapid irradiance ramps
R²	$1 - \frac{SS_{res}}{SS_{tot}}$	Dimensionless [0,1]	Proportion of variance explained; intuitive 0-1 scale; sensitive to target variance.
MAPE*	$\frac{100}{n} \sum \frac{ y_i - \hat{y}_i }{y_i}$	Percent(%)	*Undefined when $y_i = 0$; must exclude night-time observations before computing

For this dataset, a well-performing model should achieve MAE < 5kW, RMSE < 8kW, and R² < 0.97 on the 2019 test set. The MAPE should be computed only on daytime observations (GHI > 0) to avoid division by zero during night-time hours.

RQ9. Which machine learning methods are suitable for this problem and why?

Given the characteristics of this dataset – continuous target variable, strong GHI predictor, moderate non-linearity during cloudy periods, 31,610 training examples, and temporal structure – the following models are recommended in order of increasing sophistication:

Model	Priority	Strengths	Limitations	Expected R ²
Linear Regression	Baseline	Interpretable; fast; good starting benchmark	Cannot capture non-linearity; underperforms on cloudy days	0.90 - 0.93
Random Forest	Primary	Non-linear; robust; feature importance; little tuning	Cannot extrapolate; slower inference than linear	0.97 - 0.98
XGBoost/LightGBM	Primary	State-of-the-art accuracy; regularised; handles missing values	More hyperparameters; requires cross-validation tuning	0.98 - 0.99

SVR (RBF kernel)	Optional	Effective in high-dimensional spaces; margin-based	Slow on large datasets; kernel selection required	0.95 - 0.97
LSTM Neural Network	Advanced	Temporal dependencies; multi-step forecasting capability	High compute cost; complex tuning; overkill for 1-step	0.97 - 0.99

The primary recommendation is to begin with XGBoost or Random Forest. Both algorithms natively handle the non-linear relationship between GHI and PV power during partially cloudy periods, require minimal preprocessing, and provide built-in feature importance scores that confirm the physical interpretation (GHI dominance). Linear Regression serves as an indispensable baseline – any more complex model must exceed its R^2 to justify its additional training and inference cost. LSTM should be considered only when multi-step forecasting (ie, producing a 24-hour-ahead forecast as a sequence) is explicitly required.

VIII. PROPOSED MACHINE LEARNING PIPELINE

8.1 Pipeline Architecture

The end-to-end ML pipeline for this study comprises five stages: data ingestion and cleaning, feature engineering, model training and selection, hyperparameter optimization, and performance evaluation. Each stage feeds into the next without information leakage from test to training data.

8.2 Model Training Protocol

All models are trained on the 2016-2018 chronological training set. Hyperparameter optimization is performed using TimeSeriesSplit cross-validation with 5 folds, ensuring no future data leaks into any validation fold. For XGBoost, the primary hyperparameters to tune are:

- $N_estimators \in [100, 2000]$
- $Max_depth \in [3, 8]$
- $Learning_rate \in [0.01, 0.3]$
- $Subsample \in [0.6, 1.0]$
- $Colsample_bytree \in [0.6, 1.0]$

8.3 Expected Performance Benchmarks

Model	MAE (kW)	RMSE (kW)	R ² Score	Verdict
Linear Regression	8-14	12-20	0.90-0.93	Acceptable
Support Vector Regression	5-9	8-14	0.94-0.97	Good
Random Forest	3-6	5-9	0.97-0.98	Very Good
XGBoost / LightGBM	2-5	3-7	0.98-0.99	Excellent
LSTM (sequence model)	2-5	4-8	0.97-0.99	Excellent

IX. DISCUSSION

9.1 Implications of the GHI-PV Correlation

The near-perfect correlation ($r = 0.9995$) between GHI and PV Power is both a strength and limitation of this dataset. As a strength, it means that even a simple linear models with GHI as the sole predictor can achieve high accuracy ($R^2 \approx 0.999$ on training data). As a limitation, it raises the question of whether the marginal benefit of sophisticated ML models over linear regression is practically significant in this specific dataset. The read added value of non-linear models manifests during partially cloudy periods, when the GHI-PV relationship becomes non-linear due to the diffuse radiation component and inverter efficiency curves, and during transient events (rapid cloud shadows) that produce step changes in generation not well-represented by a linear mapping.

9.2 Limitations of the Study

Several limitations of the current dataset and methodology should be acknowledged. First, the dataset contains only three measurement columns, excluding potentially important variables such as wind speed (affects panel cooling), relative humidity (affects atmospheric transmittance), cloud cover fraction, and panel soiling level. Second, the dataset represents a single PV installation at an unknown location; the trained model may not generalize to different latitudes, climates, or panel technologies. Third, no uncertainty quantification is performed – probabilistic forecasting methods (e.g., quantile regression forecasts, conformal prediction) would be more useful for grid operations than point forecasts alone.

9.3 Future Directions

Several promising extensions of this work are identified. Incorporating additional meteorological variables (cloud optical depth, aerosol optical depth) from reanalysis datasets such as ERA5 would likely reduce RMSE by a further 10-20%. Probabilistic forecasting using quantile regression or Bayesian approaches would provide uncertainty bounds essential for grid planning. Transfer learning across multiple PV sites could enable models trained on data-rich sites to improve forecasts at data-sparse sites. Finally, combining physics-based models (to handle extrapolation to unseen irradiance conditions) with data-driven ML models in a hybrid architecture represents the current research frontier in the field.

X. CONCLUSIONS

This paper has presented a comprehensive investigation of solar PV power forecasting as a supervised machine learning regression problem, grounded in a real four-year hourly dataset of 31,610 observations. The following conclusions were found:

1. GHI is the dominant predictor of PV power generation, exhibiting a Pearson correlation of $r = 0.9995$ – the highest single-feature predictive relationship feasibility achievable in a real-world physical system.
2. PV Power is the appropriate target variable, casually downstream of whether inputs and operationally significant for grid management.
3. The dataset exhibits a clear bell-shaped diurnal generation profile (zero at night, peak at solar noon) and seasonal variation over the four-year span.
4. Preprocessing requirements are minimal: handling 16 missing values, parsing DateTime to engineer temporal features, applying StandardScaler, and performing a chronological 80/20 train/test split.
5. XGBoost and Random Forest are the recommended primary models, expected to achieve $R^2 > 0.97$, $MAE < 5kW$, and $RMSE < 8kW$ on the 2019 test set, substantially outperforming the Linear Regression baseline.
6. Model evaluation should use MAE, RMSE, and R^2 together to capture both average accuracy and sensitivity to large prediction errors.
7. The methodology developed in this paper – from physical background through preprocessing, modelling, and evaluation – provides a rigorous and reproducible framework applicable to any single-site solar PV forecasting problem.

As the global share of solar energy continues to grow, accurate ML-based forecasting will become an increasingly critical component of grid management infrastructure. The methods and conceptual foundations presented in this paper equip scientists with the knowledge required to design, implement, and evaluate effective solar power forecasting systems.

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